

Electrical Design

4.1 Overview

The 2026 electrical architecture is organised into four subsystems: power distribution, network and computation, sensing, and motor control. Figure ?? shows how these subsystems interconnect.

- **Power** — a 48 V battery feeds two buck converters (12 V and 5 V rails) that supply the rest of the vehicle (Section 4.2).
- **Compute** — a Jetson Orin GPU handles perception and high-level control. It commands a Teensy 4.1 microcontroller over USB serial (Section 4.5).
- **Sensing** — three ZED X stereo cameras, a Velodyne LiDAR, and an Xsens GPS feed the Jetson (Section 4.4).
- **Motor control** — the Teensy runs a closed-loop controller and commands two SPARK MAX motor controllers over CAN, which in turn drive two NEO brushless motors (Section 4.3).

The emergency-stop circuit is implemented in hardware: it directly interrupts the supply to both SPARK MAX controllers, immediately removing torque from both motors regardless of software state.

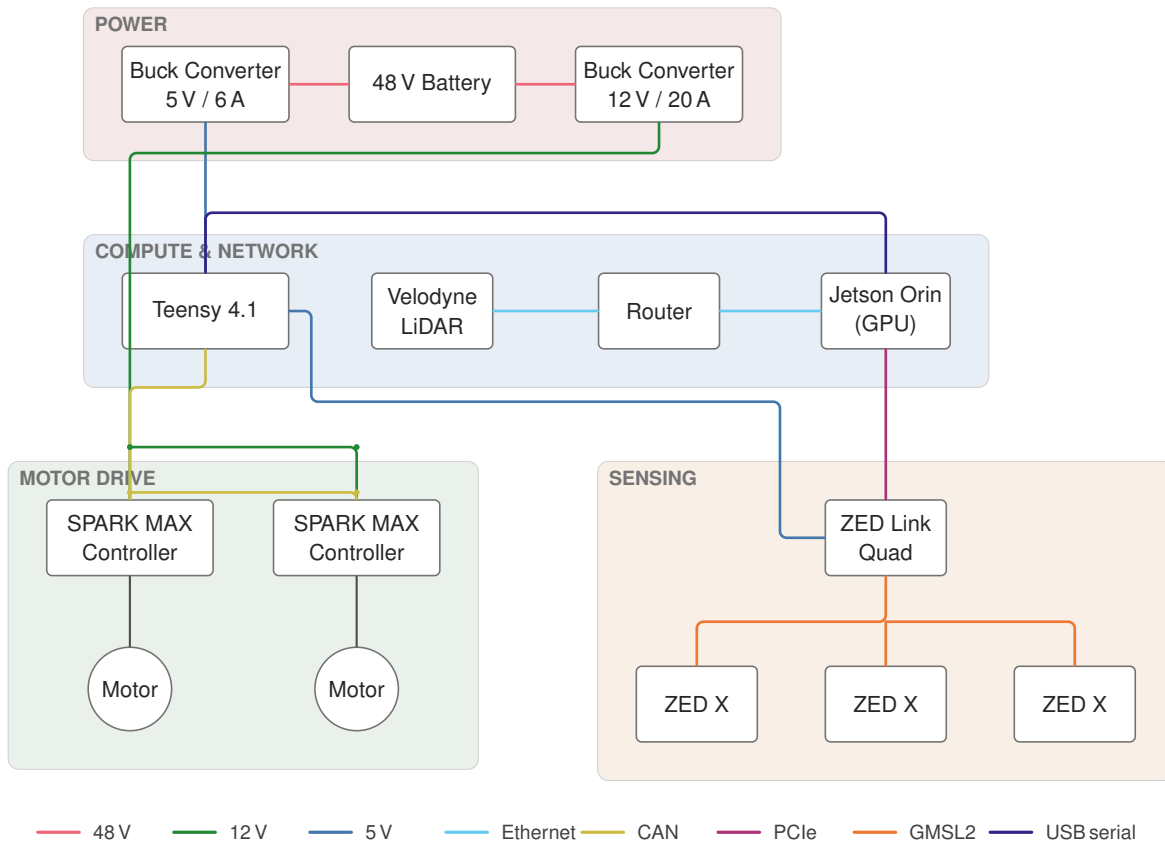


Figure 1: System-level electrical block diagram. Subsystems are grouped by shaded zones; line colour denotes bus type per the legend.

4.2 Power System

A 48 V battery is the single power source on the vehicle. As shown in Figure ??, the battery passes through a main switch into two buck converters: a 12 V / 20 A converter for the motor controllers and other high-power loads, and a 5 V / 6 A converter for the Teensy 4.1 and low-power peripherals.

Each rail is split into three independently fused output branches. Per-rail fusing isolates faults so that a short on one load does not pull down the rest of the vehicle.

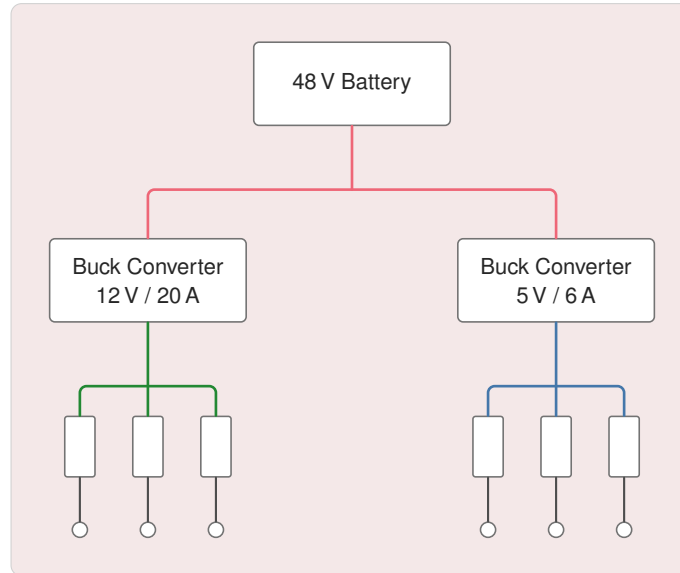


Figure 2: Power distribution from the 48 V battery through two buck converters to fused output rails. Trunk colours match the bus legend of Figure ??.

4.3 Motor Control

The drivetrain uses two NEO brushless motors, each driven by a SPARK MAX motor controller. Both controllers share a single CAN bus to the Teensy 4.1, which acts as the real-time control node. Each SPARK MAX reports position and velocity from its integrated encoder back over the same bus.

The Teensy closes a PID velocity loop per wheel, using encoder feedback from the SPARK MAX controllers and reference velocities published by the Jetson over USB serial. Because the motor controllers, the encoders, and the microcontroller all live on one CAN bus, no separate signal wiring between the motor controllers and the Teensy is required.

The hardware emergency stop interrupts the SPARK MAX supply directly. When triggered, both controllers lose power and the motors coast to a stop; software is not in the loop.

4.4 Sensing

The vehicle uses three classes of sensor:

- **Stereo cameras** — three ZED X cameras connected to a ZED Link Quad over GMSL2. The ZED Link Quad attaches to the Jetson over PCIe, keeping the cameras off the Ethernet network.
- **LiDAR** — a Velodyne LiDAR provides 3D point clouds for obstacle detection and mapping. It reports to the Jetson over Ethernet through the on-board router.
- **GPS / IMU** — an Xsens module provides position and orientation. It connects directly to the Jetson.

Camera, LiDAR, and GPS streams are time-aligned on the Jetson before being consumed by perception and localization.

4.5 Network and Computation

The Jetson Orin is the primary computation unit. It runs perception, sensor fusion, and trajectory generation, and publishes wheel velocity commands to the Teensy.

An on-board router carries Ethernet between the Jetson and the Velodyne LiDAR. The Teensy 4.1 connects to the Jetson via USB serial (115200 baud) rather than Ethernet. The high-bandwidth camera link is kept separate: the ZED Link Quad attaches directly to the Jetson over PCIe so that the three ZED X feeds do not contend for the Ethernet network. The Xsens GPS connects directly to the Jetson.

The same router also exposes a Wi-Fi link to a remote laptop for monitoring, data capture, and remote shutdown during testing.